Chapter 1

Supplementary Information for Chapter 2: Helium Step Heat Experiment on Goethite in Hydrothermal Conditions

1.1 Permeation Lag Correction and Ampoule Permeability

1.1.1 Permeation Lag Correction

Because HT-Win-06-01b was encapsulated in an imperfectly helium-transmissive silica glass ampoule, permeation lag was an expected complication in these measurements. To address this, experimental steps at temperatures at or above 170°C were bracketed by low temperature (<120°C) steps where the helium in goethite at high pH2O was effectively immobile at experimental timescales, following the permeation step method (Chapter **??**, §**??**). For 3He, measured signals at these low temperature permeation steps were entirely controlled by the process of permeation through silica glass, and therefore could be used to estimate the amount of helium remaining within the ampoule between high-temperature experimental steps. However, for 4He, the measured signals at low temperature permeation steps were most often indistinguishable from the blank, so Eq. (**??**) (Chapter **??**) could not be applied to 4He and the permeation step method could not be carried through for that isotope.

As a result, permeation lag is instead addressed using the bounding method described in Chapter **??** (§**??**), which constrains the ampoule's residual helium from above rather than measuring it. In this approach, an upper bound constraint for the helium content of the ampoule at the end of each high-temperature step was calculated using Eq. (**??**) (Chapter **??**). 3He was reduced this way as well, for consistency with 4He; because 3He can also be reduced by the permeation step method, the two reductions can be compared directly, and that comparison is used below (§1.1.3) as an internal test of the bounding method.

Both routes require the ampoule's permeation rate constant λ, which was measured during the experiment itself rather than assumed from the literature (Chapter **??**, §**??**). This in-experiment measurement is not an alternative method of handling permeation lag but a prerequisite of the permeation step method and, here, the empirical basis for choosing a conservative λ for the bounding method. It is indispensable in this experiment: the permeability of ampoules used in hydrothermal step heats cannot reasonably be assumed to match literature values for anhydrous silica glass.

Ten series of 3–4 consecutive permeation steps held at a single temperature between 90–110°C for 120-180 minutes each with 26 minutes of cooling time before being inlet into the Helix-MC-Plus^{10K} were identified as suitable for this exercise. The 3He measurements in these series were fit to Eq. (**??**) using a non-linear least squares optimization (Fig. 1.1). This is accomplished with the Levenberg-Marquardt algorithm implemented in the curve_fit function in the SciPy package of Python (Virtanen et al., 2020), weighted by each measurement's own uncertainty. One additional series that otherwise qualified was not well fit by this model and was not used further for clarity, though its fit value of λ was within uncertainty of the other series' fit values for λ.

An initial guess of $\lambda \approx 6{\times}10^{-6}$ sec^{-1} and $n_0 \approx \frac{m_i}{1-e^{-\lambda t_i}}$ (evaluated at the first measurement of a series) pccSTP was passed to the optimizer for each series. Uncertainties from the model fits were calculated using the covariance matrices generated for each of the model fits. Although the initial helium amount n_0 was a free parameter in the optimizations, the rate constant of permeation (λ) was the only parameter of interest for this exercise. The fitted rate constants span 2.7×10^{-5} to 6.3×10^{-5} s^{-1}, with 2σ uncertainties of 0.08 to 0.46 log units and no trend resolvable against step duration:

All measured apparent 3He permeabilities were equal to or in excess of the literature values for anhydrous silica glass (Fig. 1.2), similar to observations discussed in Chapter **??** (§**??**). 4He permeability itself could not be independently measured in this experiment due to blank-level signals during the low temperature permeation steps and is assumed to similarly exceed the literature values for anhydrous silica glass. To evaluate the uncertainty due to the permeation lag, we use the literature values as a conservative (lower bound) estimate of the true permeability; a λ that underestimates the ampoule's true rate constant yields a conservative (higher) upper bound on the residual helium (Chapter **??**, §**??**). λt (where $\lambda = GRTK$) is estimated here using the 4He permeability K reported by Swets et al. (1961), while the 3He permeability is taken to be the product of the 4He permeability and the relative 3He/4He permeability reported by Jones (1953). The geometric constant G is calculated using the measured inner and outer diameter of the ampoule (Fig. **??**).

The first three experimental steps had values of λt lower than 4, and therefore the upper bound of the helium content of these steps could not be determined. These measurements are omitted from further analysis. Later experimental steps with values of λt lower than 4 could be determined using Eq. (**??**) from the supplement to Chapter **??** because upper bounds on previous steps' residual helium content could be evaluated.

1.1.2 Measurement and Numerical Simulation of Apparent Permeability of Experimental Ampoule

The apparent permeability values used in the numerical simulation (§1.2.1) are taken from the literature (Swets et al., 1961; Jones, 1953) and explicitly apply to an ampoule without water. However, the purpose of these simulations was simply to quantify the excess apparent permeabilities that would be observed for ampoules with a cooling history similar to that of the hydrothermal experiments. Because the predicted offset is universally small and a weak function of temperature or permeability, these simulations demonstrate that the much larger experimentally observed permeability is not an artifact of the cooling history of the hydrothermal experiments, reaffirming the conclusion that the hydrothermal experiments are indeed more permeable than the vacuum experiments.

The difference in log permeability between the actual and apparent permeabilities of the anhydrous simulation was significantly smaller than nearly all the 2σ uncertainties of the apparent permeabilities determined above for the hydrothermal Win-06-01b experiment, which ranged from

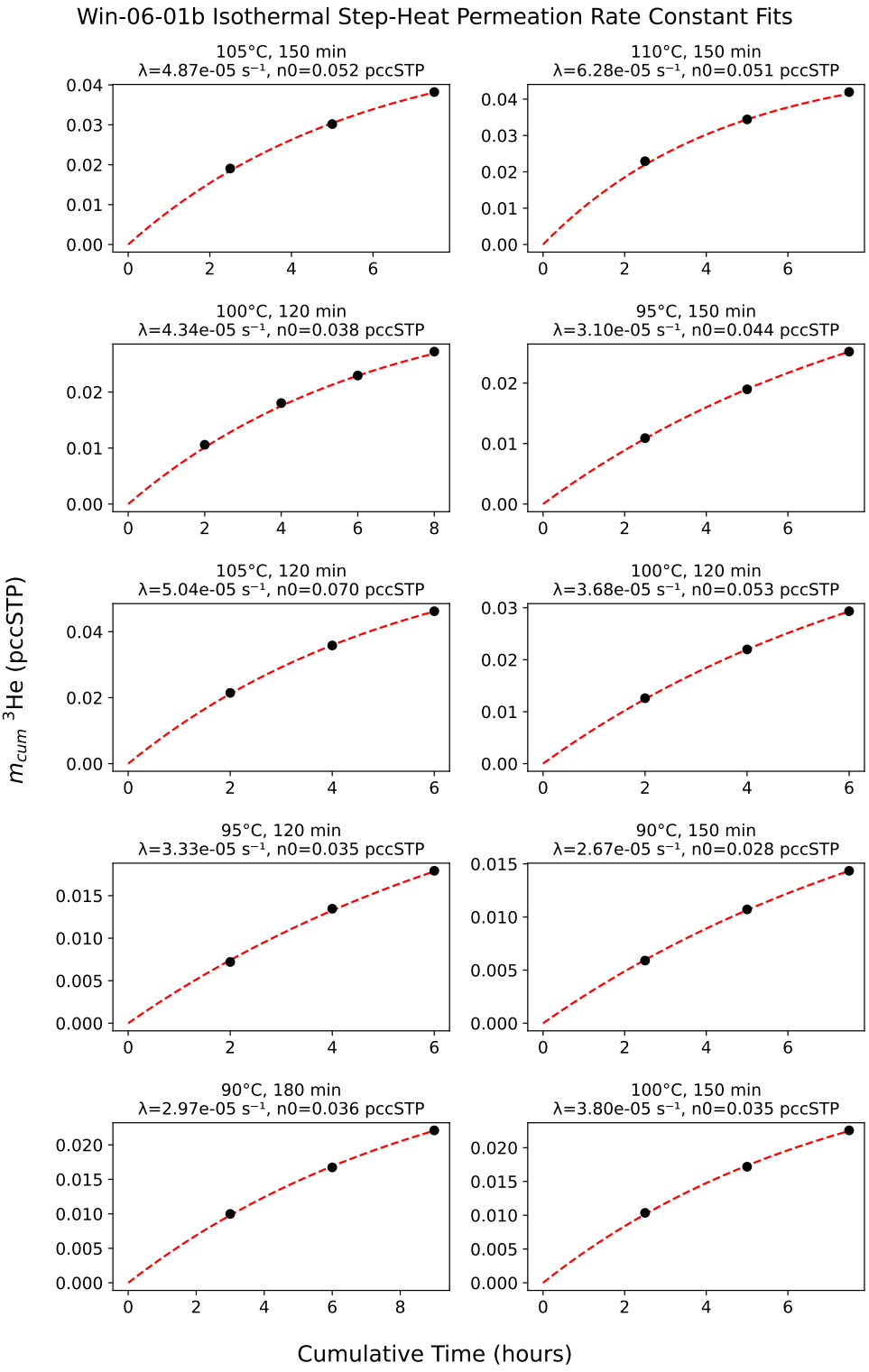


Figure 1.1: Cumulative yield against cumulative time for helium-3 yields for isothermal permeation steps. A non-linear least squares optimization fits λ and n_0 of Eq. (**??**) (Chapter **??**) to data from a series of permeation steps like these. The fitted λ is plotted as $\ln(\lambda/T)$ against inverse temperature in Fig. 1.2. The curvature of each series is what conditions the fit: a series in which the permeation rate has not yet slowed appreciably constrains λ and n_0 only weakly.

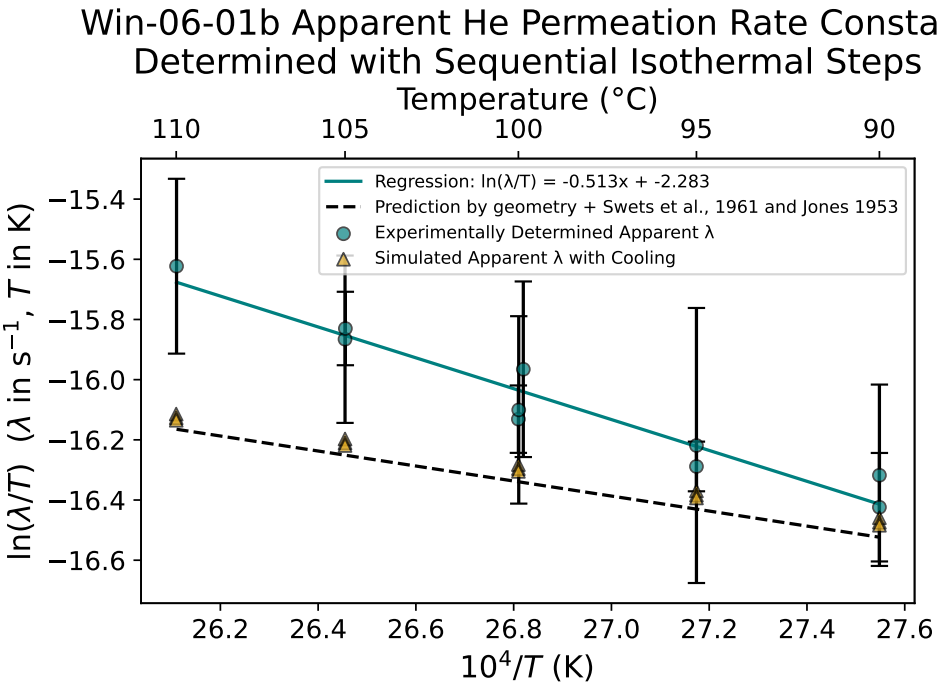


Figure 1.2: Arrhenius plot of the apparent permeation rate constant, $\ln(\lambda/T)$, as a function of inverse temperature for HT-Win-06-01b, using the rate constants fitted in Fig. 1.1. The dashed line is the prediction from the ampoule geometry and the literature permeabilities of Swets et al. (1961) and Jones (1953), converted to a rate constant via $\lambda/T = GRK$. The activation energy is higher than anhydrous silica glass, as expected (Masaryk, 1968; Shelby, 1972). It is unexpected that the glass ampoule of HT-Win-06-01b has a higher apparent 3He rate constant than predicted from the literature values of Swets et al. (1961) and Jones (1953), but this can be explained by an increase in the partial pressure of the gas phase within the tube due to displacement by high He diffusivity, low He solubility water. Yellow points labeled "Diffusion Model with cooling, Anhydrous" were produced to evaluate the expected excess apparent permeability due to the 25 minute cooldown period before measurement. This is accomplished by simulating the measured apparent permeability for a dry silica glass ampoule that continued to permeate helium while cooling for an additional 26 minutes before measurement after being held at 90–110°C for 120–180 minutes, similarly to the measured values. Variability in these points is due to differences in the length of time the ampoule was heated but is very small. They show a near constant offset from the expected values of Swets et al. (1961) and Jones (1953) that is much smaller than the uncertainty in the regression of the fit values determined experimentally in teal.

0.08 to 0.46 log units. Thus, no trends with step duration could be identified in the effective permeability values.

Therefore, the literature values underestimate the helium permeabilities of the ampoule used in this experiment, which is permissible because these values are used to calculate an upper bound constraint of the helium content of the ampoule, and lower permeabilities produce more conservative (higher) upper bound constraints.

A linear regression of $\ln(K)$ against inverse temperature (Fig. 1.2) fit to these isothermal 3He permeation measurements over 90-110°C yields an R^2 of 0.936 and an activation energy of 42.6±7.9 kJ/mol (2 s.e.), significantly higher than expected for pure silica glass with or without water (Swets et al., 1961; Jones, 1953; Masaryk, 1968; Shelby, 1972), but similar to glasses with a high abundance of network-modifying cations such as Na, K, Pb, or Li (Norton, 1953; Altemose, 1961). Because 4He signals during these isothermal steps were blank-level, only the ampoule's 3He permeability could be measured directly this way, using the fitting procedure described above.

Every 3He permeability measurement for HT-Win-06-01b equals or exceeds the anhydrous literature value, which runs counter to expectations for a hydrothermal experiment: water and hydroxyl groups dissolved in silica glass are reported to *decrease* helium permeation at all temperatures (Tsuru et al., 2011; Shelby, 1972; Masaryk, 1968), under the common paradigm that species such as $(OH)^-$ occupy interstitial space within the amorphous glassy network (Norton, 1953; Shackelford, 1999). The offset is nonetheless real: it cannot be accounted for by the ampoule's extended permeation during cooling before measurement (§1.2.1), and a similarly elevated permeability was independently observed at 100°C in a separate silica glass ampoule at Caltech partially filled with MilliQ water and equilibrated with ~0.1 torr of tank helium, as discussed in Chapter **??** (§**??**).

The likely explanation is geometric. Helium is only sparingly soluble in water and diffuses through it rapidly, roughly 7×10^{-5} cm^2/s at room temperature (Verhallen et al., 1984) against about 2×10^{-8} cm^2/s in silica glass (Swets et al., 1961). On the timescale of permeation through the glass wall, transport through the water is effectively instantaneous, so the gas phase stays in equilibrium with the glass over the entire inner surface of the ampoule, including where liquid water contacts it. Helium partitions strongly into the gas phase over both water and glass: helium concentrations in the glass and water are about 3% and 1%, respectively, of the gas-phase concentration, using solubilities from Sander (2015) and Swets et al. (1961). Displacing gas-phase volume with water therefore raises the helium partial pressure without changing the surface area available for permeation, and because the rate of permeation is proportional to the gas-phase partial pressure (Norton, 1953; Altemose, 1961; Swets et al., 1961; Masaryk, 1968), the ampoule's rate constant scales with the inverse volume fraction of the gas phase (Chapter **??**, Eq. (**??**)), generalized here to any condensed payload: $\lambda_{ampoule} = \lambda_{empty}\left(V_{condensed} + V_{gas}\right)/V_{gas}$. This observed complexity emphasizes the need to measure the ampoule's rate constant in-experiment as described in Chapter **??** (§**??**).

1.1.3 Consistency of the Bounding Method with Direct Permeation-Step Quantification: The Complete Degas Test

All Win-06-01b step heat data presented in this chapter are reduced using the bounding method (Eq. (**??**)), because 4He permeation-step yields were near blank level and could not support the permeation step method directly. As an internal check on this choice, the ampoule's 3He inventory was also reconstructed directly from the permeation steps themselves, using the in-experiment rate constants of Fig. 1.2 (§1.3). Reducing 3He by the bounding method gives Stage 1 activation energies between 157.1 and 160.6 kJ/mol and closure temperatures between 97.5 and 101.2°C across its lower-bound, central, and upper-bound reductions, a spread comparable to the regression uncertainty

itself. Fig. 1.3 compares the two: n_t, the ampoule's residual helium quantified directly from a permeation step, against $n_{t,SS}$ (equivalent to $n_{t,\mathrm{UB}}$ used elsewhere in this chapter), the upper bound on that same residual calculated independently from the preceding high-temperature step. For every ordinary permeation step in the experiment, $n_t \leq n_{t,SS}$, exactly as expected if the steady-state approximation genuinely bounds the ampoule's residual helium from above. This agreement is the justification for reducing the 4He dataset by the bounding method alone, since no analogous direct check of 4He is possible.

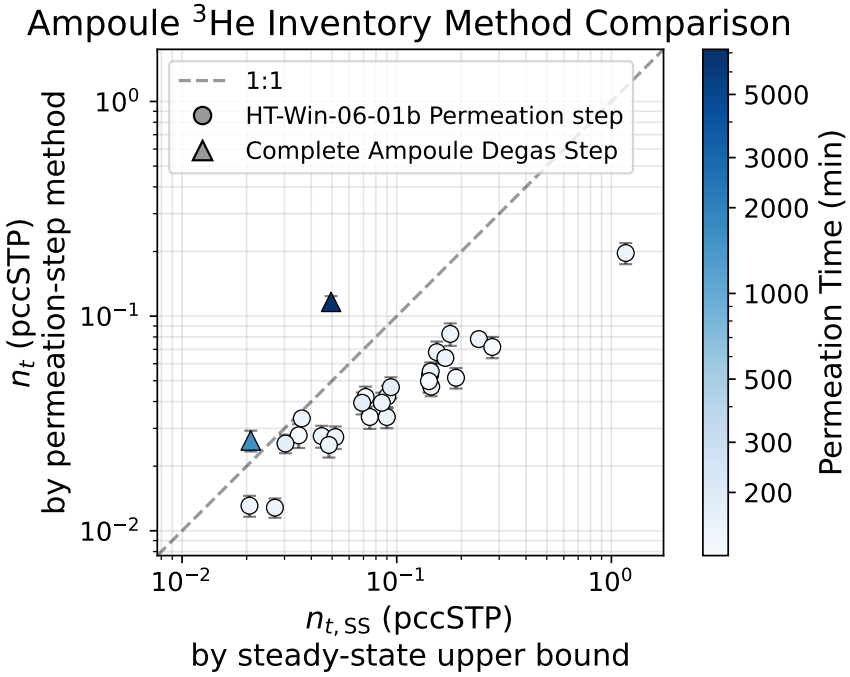


Figure 1.3: Two independent estimates of the ampoule's residual 3He inventory, n_t, following each permeation step of HT-Win-06-01b: quantified directly from the permeation step itself (*permeation step method*, y-axis, §1.3), against the upper bound calculated independently from the preceding high-temperature step (*bounding method*, x-axis, Eq. (**??**)). Color indicates the duration of the permeation step. Circles are ordinary steps; triangles are the two "attempted complete degas" steps, held at 100°C for far longer than any other permeation step (1580 and 7200 minutes) in an attempt to drive the ampoule's 3He inventory toward zero. The dashed line is 1:1. Excepting the two complete-degas steps, n_t never exceeds $n_{t,SS}$, as required if permeation lag is genuinely bounded above by the steady-state approximation. Both complete-degas steps instead record more helium than their own $n_{t,SS}$ permits.

Beyond the ampoule's residual helium content itself, the two methods' derived apparent diffusivities agree just as closely (Fig. 1.5, §1.3): $\ln(D/a^2)$ from the permeation step method tracks the bounding method's central value along the 1:1 line within uncertainty across the full range of the experiment. Fitting the permeation step method's own diffusivities with the same Stage 1 procedure used for the bounding method in the main text (Fig. 1.6, §1.3) yields an activation energy of 157.0±4.8 kJ/mol (R^2=0.991, n=12) and a closure temperature of 97.5±7.8°C, indistinguishable from the bounding method's own 157.1±5.3 kJ/mol and 98.1±8.6°C (Fig. **??**).

The two "complete degas" permeation steps depart from this pattern. For the longer of the two (7200 minutes; predicted $n_t/n_0 \approx 1 \times 10^{-8}$ under simple first-order decay), the directly measured 3He inventory exceeds $n_{t,SS}$ by a factor of 1.5. For the shorter (1580 minutes; predicted $n_t/n_0 \approx 0.02$), the excess is smaller, 25% over $n_{t,SS}$. A measured inventory in excess of $n_{t,SS}$ requires that the source term (Ω: §**??**) increased during the high temperature experimental step, or the goethite sample was not effectively closed to helium loss during the low temperature permeation step (§**??**), or both. The measured diffusivity is constant at a given temperature for these high temperature measurements at the end of the experiment (Figs. **??**, **??**), so there is no evidence the source term (Ω) is increasing during the high temperature measurements.

The size of the two excesses scales with the duration of their respective permeation step. The 7200-minute step is associated with the larger (1.5×) excess whereas the 1580-minute step is associated with the smaller (25%) excess. An artifact intrinsic to the first-order process of helium permeation through the ampoule itself, such as an underestimated rate constant, has no reason to strongly scale with the step duration, since both of these long steps would be expected to empty the vast majority of the helium inventory of the ampoule. That the excess does scale this way points to open system behavior in the sample. The preferred explanation is short-circuit or fast pathway diffusion (Lee and Aldama, 1992) of residual helium held within the interstitial space of the goethite crystallite aggregate (§**??**), recharged during the preceding high-temperature step, released slowly and with weak temperature dependence during the following extended low-temperature hold. This requires communication between a lattice or grain-boundary source and an interstitial reservoir, a departure from the independent, non-communicating domains of a strict MDD interpretation, but it is consistent with the interstitial framework already invoked to explain the early, low-retention fraction of 3He release (§**??**). Unlike dehydration, it requires no change in the intrinsic retentivity of the crystalline lattice recorded by the bulk of the experiment.

1.2 Analytical Methods

1.2.1 Finite Difference Diffusion Model of Cooling Tube

One potential concern about the results of the internally measured permeabilities is that some additional helium was permeated during the cooling period after each permeation step. In order to evaluate the significance of this effect, a finite difference model was built in Python to simulate diffusion, influx, and outflux of helium across a hollow glass cylinder containing an arbitrary amount of helium gas within a helium-free chamber using the procedure described in Chapter **??** (§**??**). The glass cylinder was given the same dimensions of the ampoule used in the Win-06-01b hydrothermal experiment, and was subjected to the same heating and cooling schedule as the hydrothermal experiment's permeation steps. The tube was charged with 100 arbitrary units of helium atoms at the end of each experimental step and a steady state concentration profile was assumed for the new arbitrary charge at T = 24°C. Permeability values for 3He were assumed as the product of the 4He permeability values given by Swets et al. (1961) and the 3He/4He relative permeabilities given by Jones (1953).

To simulate cooling, a cooling rate had to be assumed. To most accurately simulate the cooling of the experimental ampoule, Watlow temperature records of the experimental ampoule during cooling steps were used to calculate the cooling rate dT/dt for discrete time intervals. A quadratic function was fit to these data points to model the cooling rate as a function of temperature with a single root at T = 24°C such that the temperature would stabilize at T = 24°C by design.

$$\frac{dT}{dt} = -\left(9.23081 \times 10^{-6}\right)(T-24)^2 - \left(7.74304 \times 10^{-4}\right)(T-24)$$

(dT/dt in °C/s, T in °C; $R^2 = 0.9633$).

1.2.2 Uncertainty for Diffusivity Inversions

Apparent values of D/a^2 for 3He and 4He were calculated from times and cumulative measured yields using the equations of Fechtig and Kalbitzer (1966). Lower and upper bounds for diffusivity were calculated using the upper bound for the helium contents of the ampoules before and after each step, evaluated using Eq. (**??**). Diffusivity for step i is a function of the cumulative fractional yield for step i $\Sigma^i F$ and the cumulative fractional yield of the previous step $\Sigma^{i-1} F$. The diffusivity is maximized when the difference between the fractional yields is largest, and minimized when the difference is smallest. This difference is maximized for the extreme case where the previous step ended with no permeation lag, and the current step ended with the maximum amount of permeation lag ($^i_{SS}n_t$), whereas the opposite is true for the minimum diffusivity.

$$\begin{aligned} {}^{UB}D_\omega &= D\left(\Sigma^{i-1}F_m,\ \Sigma^i F_m + {}^i_{SS}n_t\right) \\ {}^{LB}D_\omega &= D\left(\Sigma^{i-1}F_m + {}^{i-1}_{SS}n_t,\ \Sigma^i F_m\right) \end{aligned} \tag{1.1}$$

For the 3He diffusivities, a linear regression was fit to the steps with cumulative 3He fractional yield ≥7% (12 steps, Fig. **??**), yielding an R^2 statistic of 0.989. The fit's slope and intercept are used as a reference line to calculate "Delta" values for the diffusivities of 3He and 4He (Fig. **??**). No uncertainty on the total amounts of 3He and 4He is propagated into the diffusivities because the experiment operated at low cumulative yields where this error affects all diffusivities approximately homogeneously, so it is only relevant for the y-intercept of the regression and the corresponding calculated closure temperature. The uncertainty from the intercept and slope parameters of the

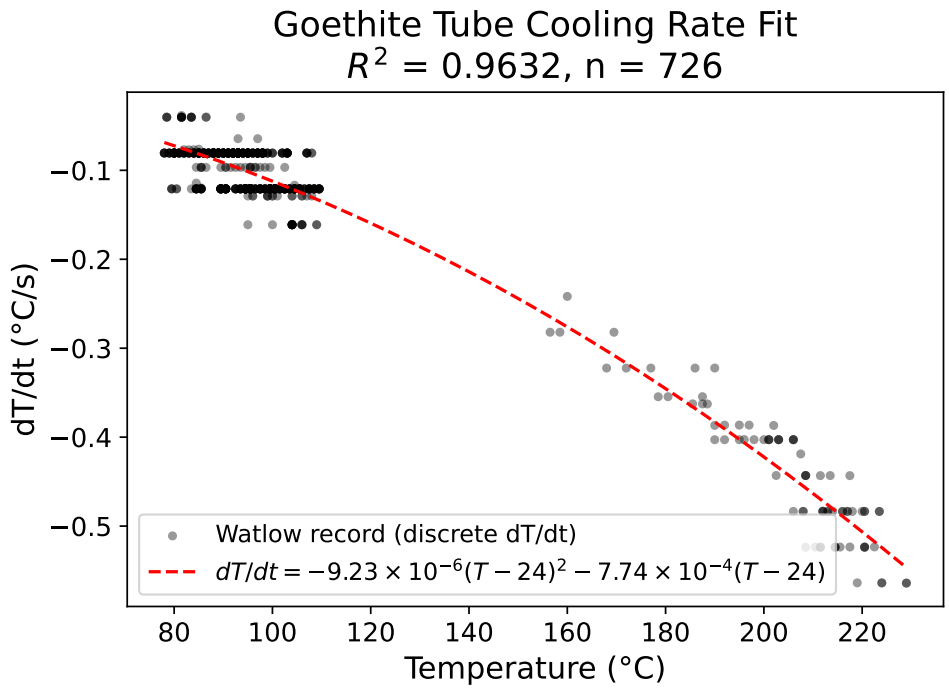


Figure 1.4: Discrete cooling rate (dT/dt) values calculated from the Watlow temperature record for the Winsor Goethite experiment, plotted against temperature, with the quadratic fit used to simulate cooling in the finite difference diffusion model overlain.

linear regression is significantly larger than the uncertainty related to the total yield, so it is neglected.

1.2.3 Helium Isotope Ratio Uncertainty

The upper and lower bounds of the $\frac{^4He}{^3He}$ isotope ratios for each heating step are taken as the maximum and minimum $\frac{^4He}{^3He}$ isotope ratio calculated out of the 4 hypothetical end-member permeation scenarios, which would have been experienced by both helium isotope systems:

1. $n_t = 0,\ n_0 = 0$
2. $n_t = n_t^{\mathrm{UB}},\ n_0 = 0$
3. $n_t = n_t^{\mathrm{UB}},\ n_0 = n_0^{\mathrm{UB}}$
4. $n_t = 0,\ n_0 = n_0^{\mathrm{UB}}$

Intuitively, higher differential permeability between the two helium isotopes ($\delta = \frac{K_{3He}}{K_{4He}}$ here) will result in greater isotope fractionation between the measured $\left(\frac{m_4}{m_3}\right)$ and true $\left(\frac{\omega_4}{\omega_3}\right)$. Furthermore, lower absolute permeabilities will result in stronger permeation lag, additionally producing higher isotope fractionation in the measured value. The 4He permeability is thus conservatively taken to be identical to Swets et al. (1961), an anhydrous literature value which this experiment's own isothermal 3He permeation measurements show to underestimate the ampoule's true permeability (Fig. 1.2); 4He permeability itself could not be independently measured in this experiment. For the purposes of evaluating the measured isotope ratios, the relative 3He/4He permeability (δ) is taken to be 1.29, the 2σ upper bound of the value measured independently for a water-filled ampoule in Chapter **??** (§**??**): 1.10 ± 0.19 (2σ). This value of 1.29 far exceeds realistic values of the relative 3He/4He permeability, which is expected to be between 1.06 and 1.11 for the temperature range relevant to this experiment (Jones, 1953). These conservative assumptions taken together guarantee that the calculated upper and lower bounds for the isotope ratios accurately constrain the true value. The upper bounds of the ampoule contents are calculated using the steady-state approximation of Eq. (**??**) (Chapter **??**): $n_{ss} = \frac{m}{\lambda t - 1}$. For the four extreme cases listed above, the following equations apply:

1. $$\frac{\omega_4}{\omega_3} = \frac{m_4}{m_3}$$
2. $$\frac{\omega_4}{\omega_3} = \frac{m_4 + \frac{m_4}{\lambda t - 1}}{m_3 + \frac{m_3}{\delta \lambda t - 1}}$$
3. $$\frac{\omega_4}{\omega_3} = \frac{m_4 + \frac{m_4}{\lambda t - 1} - \frac{m_4^{i-1}}{(\lambda t)^{i-1} - 1}}{m_3 + \frac{m_3}{\delta \lambda t - 1} - \frac{m_3^{i-1}}{\delta (\lambda t)^{i-1} - 1}}$$
4. $$\frac{\omega_4}{\omega_3} = \frac{m_4 - \frac{m_4^{i-1}}{(\lambda t)^{i-1} - 1}}{m_3 - \frac{m_3^{i-1}}{\delta (\lambda t)^{i-1} - 1}}$$

Where m_x^{i-1} and $(\lambda t)^{i-1}$ refer to the measured helium content and λt of the previous step, respectively. λ here applies explicitly to 4He, whereas $\delta\lambda$ applies to 3He. The upper and lower bounds of the isotope ratios are then taken to be the maximum and minimum of these four extreme cases.

1.3 Goethite Permeation Rate Constant Dataset

The in-experiment permeation rate constants used to correct for permeation lag are reported below. Table 1.1 gives the raw per-step data for all 10 isothermal sequences, and Table 1.2 gives the lambda/n_0 fit to each sequence (Fig. 1.1) that together define the Arrhenius regression of Fig. 1.2.

Table 1.1: Raw step-heat data ('Fit Lambdas' sheet, cross-checked against 'Raw Data and Corrections') for the 10 isothermal 3He permeation sequences used to determine the ampoule's apparent permeability (Fig. 1.1, Fig. 1.2). Series are separated by horizontal rules; within a series, steps are listed in the order measured.

Series	Temp (°C)	Nominal Time (min)	Cooling Time (min)	^{3}He measured (pccSTP)	$\pm 2\sigma$
1	105.00	150	26	0.01905	0.00011
1	105.00	150	26	0.011134	0.000074
1	105.00	150	26	0.008027	0.000053
2	109.99	150	26	0.02292	0.00014
2	110.00	150	26	0.011512	0.000073
2	110.00	150	26	0.007518	0.000052
3	99.86	120	26	0.010584	0.000061
3	100.00	120	26	0.007448	0.000040
3	100.00	120	26	0.004916	0.000025
3	99.99	120	26	0.004263	0.000021
4	95.00	150	26	0.01090	0.00028
4	95.00	150	26	0.00807	0.00020
4	95.00	150	26	0.00622	0.00014
5	104.99	120	26	0.021450	0.000089
5	104.99	120	26	0.014372	0.000060
5	104.99	120	26	0.010381	0.000043
6	100.00	120	26	0.012600	0.000056
6	100.00	120	26	0.009377	0.000035
6	100.00	120	26	0.007355	0.000020
7	95.00	120	26	0.007203	0.000030
7	95.00	120	26	0.006262	0.000026
7	95.00	120	26	0.004461	0.000019
8	90.00	150	26	0.005899	0.000027
8	89.90	150	26	0.004812	0.000024
8	90.00	150	26	0.003635	0.000019
9	90.00	180	26	0.009987	0.000062
9	89.92	180	26	0.006748	0.000051
9	90.00	180	26	0.005370	0.000048
10	99.99	150	26	0.010350	0.000043
10	99.98	150	26	0.006832	0.000035
10	100.00	150	26	0.005377	0.000034

Table 1.2: Fitted λ (permeation rate constant) and n_0 (residual helium at the start of the series) for each of the 10 isothermal sequences in Table 1.1, from the weighted non-linear least-squares fit of Eq. (**??**) (Chapter **??**) described in §1.1. n is the number of steps in the sequence. These are the same per-sequence fits plotted in Fig. 1.1 and used to build the Arrhenius regression in Fig. 1.2.

Series	Temp (°C)	Duration (min)	n	λ (s^{-1})	$\pm 2\sigma$	n_0 (pccSTP)	$\pm 2\sigma$
1	105.00	150	3	4.9×10^{-5}	1.4×10^{-5}	0.0521	0.0078
2	109.99	150	3	6.3×10^{-5}	1.8×10^{-5}	0.0508	0.0073
3	99.86	120	4	4.3×10^{-5}	1.3×10^{-5}	0.0377	0.0052
4	95.00	150	3	3.10×10^{-5}	2.6×10^{-6}	0.0444	0.0022
5	104.99	120	3	5.04×10^{-5}	6.2×10^{-6}	0.0696	0.0047
6	100.00	120	3	3.68×10^{-5}	4.1×10^{-6}	0.0534	0.0034
7	95.00	120	3	3.3×10^{-5}	1.5×10^{-5}	0.035	0.011
8	90.00	150	3	2.67×10^{-5}	4.8×10^{-6}	0.0279	0.0035
9	90.00	180	3	2.97×10^{-5}	9.0×10^{-6}	0.0357	0.0070
10	99.99	150	3	3.8×10^{-5}	1.2×10^{-5}	0.0351	0.0070

^{3}He Step Heat Dataset, Corrected with the Permeation Step Method

The complete corrected step-heat dataset produced by the permeation step method, using the in-experiment rate constants above (Fig. 1.2), is reported in Table 1.3, giving the central value and 2σ uncertainty for each quantity, analogous to Table **??** in the main text for the bounding method. 4He is not treated by the permeation step method here: 4He permeation-step signals were blank level, so no direct, step-by-step quantification of the ampoule's 4He inventory was possible, and 4He throughout this chapter is instead reduced exclusively by the bounding method of the main text (§1.2). The permeation step dataset reported here is retained only for 3He, as the internal check on the bounding method discussed in §1.1.3.

Table 1.3: ^{3}He step-heat dataset for HT-Win-06-01b under the permeation step method: measured and cumulative ^{3}He, the initial and residual ampoule content (n_0, n_t), the true step production (ω), and the apparent diffusivity $\ln(D/a^2)$, each with its 2σ uncertainty. Six rows have derived values marked with "*": these steps have measured ^{3}He < 0.1 pccSTP and are excluded as low-signal, as in Table **??**.

Temp (°C)	Time (min)	^{3}He meas. (pccSTP)	Cum. ^{3}He (pccSTP)	n_0 (pccSTP)	$\pm 2\sigma$	n_t (pccSTP)	$\pm 2\sigma$	ω (pccSTP)	$\pm 2\sigma$	$\ln(D/a^2)$	$\pm 2\sigma$
170	300	0.9630	0.0000	0	0	0.197	0.022	1.160	0.026	-21.910	0.045
180	463	0.5442	0.9630	0.073	0.013	0.0782	0.0049	0.550	0.016	-22.184	0.089
190	550	0.5292	1.6241	0.0195	0.0025	0.0679	0.0082	0.578	0.013	-21.976	0.085
200	500	0.5190	2.2112	0.0328	0.0054	0.0532	0.0058	0.539	0.012	-21.702	0.088
210	475	0.5770	2.7646	0.0141	0.0021	0.0552	0.0057	0.618	0.012	-21.311	0.082
220	450	0.6333	3.3799	0.0110	0.0018	0.0469	0.0046	0.669	0.012	-20.991	0.080
230	425	0.7899	4.0551	0.0059	0.0019	0.0639	0.0035	0.848	0.012	-20.514	0.066
225	908	0.7513	4.8885	0.0195	0.0024	0.0420	0.0049	0.774	0.014	-21.20	0.12
215	450	0.1415	5.6842	0.0131	0.0018	0.0278	0.0035	0.1563	0.0065	-22.02	0.32
205	520	0.0763	5.8529	0.0072*	0.0012*	0.0125*	0.0016*	0.0816*	0.0081*	-22.79*	0.83*
195	1106	0.0670	5.9491	0.00212*	0.00040*	0.0307*	0.0029*	0.0956*	0.0032*	-23.4*	1.1*
185	740	0.0227	6.0246	0.0125*	0.0016*	0.0090*	0.0011*	0.0192*	0.0020*	-24.6*	2.8*
175	920	0.0180	6.0610	0.00368*	0.00050*	0.0045*	0.0004*	0.0189*	0.0007*	-24.8*	3.8*
200	260	0.0224	6.0814	0.00065*	0.00012*	0.0095*	0.0012*	0.0313*	0.0016*	-23.0*	1.5*
210	520	0.0812	6.1081	0.0075*	0.0010*	0.0145*	0.0013*	0.0882*	0.0038*	-22.7*	1.1*
220	1167	0.5766	6.1914	0.0028	0.0014	0.0276	0.0032	0.601	0.024	-21.50	0.18
230	370	0.3577	6.7796	0.0087	0.0040	0.0422	0.0051	0.391	0.015	-20.71	0.15
240	320	0.6767	7.1563	0.0172	0.0024	0.0826	0.0099	0.742	0.016	-19.807	0.099
245	250	0.5549	7.8582	0.0213	0.0089	0.0516	0.0056	0.585	0.015	-19.71	0.12
250	210	0.6934	8.4743	0.0108	0.0045	0.0718	0.0082	0.754	0.014	-19.20	0.10
245	270	0.4659	9.2086	0.0248	0.0035	0.0498	0.0058	0.491	0.010	-19.81	0.20
240	320	0.2857	9.7206	0.0207	0.0028	0.0339	0.0041	0.2989	0.0066	-20.43	0.31
235	370	0.2199	10.0357	0.0166	0.0022	0.0273	0.0033	0.2307	0.0052	-20.81	0.40
230	420	0.1400	10.2736	0.0132	0.0018	0.0255	0.0025	0.1523	0.0040	-21.33	0.60
225	652	0.1494	10.4279	0.0126	0.0016	0.0131	0.0015	0.1499	0.0034	-21.77	0.75
230	500	0.1550	10.5908	0.0007	0.0017	0.0128	0.0013	0.1671	0.0034	-21.38	0.46
235	450	0.1951	10.7582	0.00257	0.00042	0.0334	0.0018	0.2259	0.0033	-20.96	0.36
240	400	0.2425	10.9638	0.0069	0.0014	0.0251	0.0032	0.261	0.012	-20.67	0.32
245	350	0.3150	11.2328	0.0058	0.0010	0.0395	0.0046	0.3487	0.0094	-20.22	0.24
250	300	0.3716	11.5689	0.0165	0.0022	0.0467	0.0052	0.4018	0.0096	-19.89	0.21
250	300	0.3640	11.9627	0.0243	0.0044	0.0339	0.0038	0.3735	0.0080	-19.93	0.24

continued on next page

(continued)

Temp (°C)	Time (min)	^{3}He meas. (pccSTP)	Cum. ^{3}He (pccSTP)	n_0 (pccSTP)	$\pm 2\sigma$	n_t (pccSTP)	$\pm 2\sigma$	ω (pccSTP)	$\pm 2\sigma$	$\ln(D/a^2)$	$\pm 2\sigma$
250	300	0.3452	12.3490	0.0113	0.0016	0.0395	0.0045	0.3733	0.0079	-19.90	0.24
250	10250	8.4829	12.7167	0.0191	0.0025	0.1165	0.0074	8.58	0.13	-19.96	0.19
240	785	0.2261	21.2187	2.9×10^{-9}	6.8×10^{-9}	0.0263	0.0029	0.252	0.022	-20.66	0.58
Total			146.1483								

1.3.1 Permeation Step Approach and Concordance with Bounding Approach

Beyond the ampoule's residual helium content itself (§1.1.3), the two methods' derived apparent diffusivities also agree closely. Fig. 1.5 compares $\ln(D/a^2)$ from the permeation-step method against the bounding method's central value for every step in the experiment, with the bounding method's own lower- and upper-bound spread on one axis and the permeation step method's propagated uncertainty on the other. The two diffusivity datasets track the 1:1 line within uncertainty across the full range of the experiment. Fitting the permeation-step method's own diffusivities with the same Stage 1 procedure used for the bounding method in the main text (Fig. 1.6) yields an activation energy of 157.0±4.8 kJ/mol (R^2=0.991, n=12) and a closure temperature of 97.5±7.8°C, indistinguishable from the bounding method's own 157.1±5.3 kJ/mol and 98.1±8.6°C (Fig. **??**, main text).

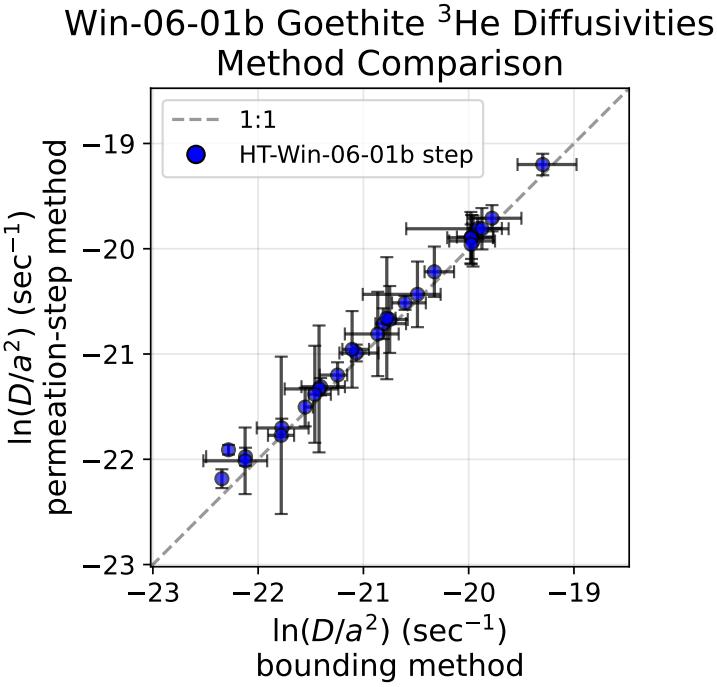


Figure 1.5: Apparent ^{3}He diffusivities for HT-Win-06-01b from the permeation-step method (y-axis) against the bounding method's central value (x-axis, Eq. (**??**)), with the bounding method's lower/upper-bound spread and the permeation step method's own propagated uncertainty plotted on their respective axes. The dashed line is 1:1.

1.4 Kinetic mechanisms and the limits of a single heating schedule

Solid-state rate laws separate the temperature dependence of the rate from its dependence on the extent of reaction (Vyazovkin et al., 2011):

$$\frac{d\alpha}{dt} = A \exp\left(-\frac{E_a}{RT}\right) h(\alpha), \tag{1.2}$$

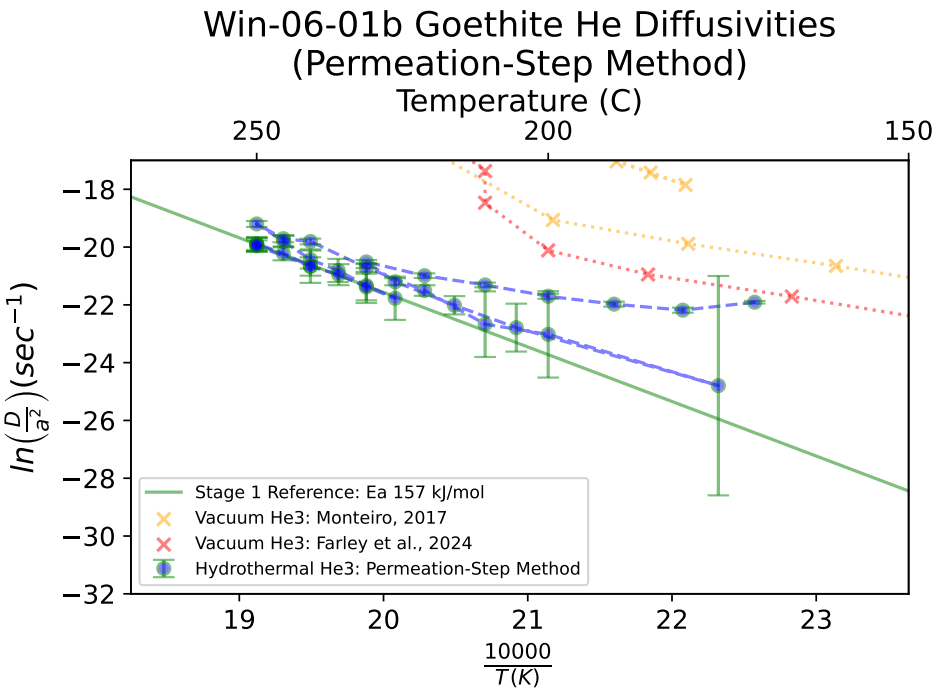


Figure 1.6: Analogue of Fig. **??** (main text) using the permeation-step method's own ^{3}He diffusivities and its own Stage 1 linear regression, rather than the bounding method's. The resulting activation energy and closure temperature (157.0±4.8 kJ/mol; 97.5±7.8°C) agree closely with the bounding method's own (157.1±5.3 kJ/mol; 98.1±8.6°C).

where α is the fraction of goethite converted. The reaction model $h(\alpha)$ expresses the mechanism: it describes how the rate changes as conversion proceeds, which depends on where and how the reaction advances through a grain. For first-order kinetics (F1), grains convert independently once nucleated, so the rate is proportional to the goethite remaining, $h = 1 - \alpha$. For a reaction front moving inward from the grain surface (R2, R3), the rate is proportional to the shrinking area of the unreacted core. For random nucleation followed by growth (A2, A3), the rate first accelerates and then slows. For any heating schedule, each rate law gives α as a function of a single variable,

$$x = A \sum_j \Delta t_j \exp\left(-\frac{E_a}{RT_j}\right), \tag{1.3}$$

the Arrhenius-weighted time delivered by the steps up to the current one; for a single isothermal step, $x = kt$. We tested the mechanisms in Table 1.4, including spherical diffusion on the same footing: for diffusion, x is the Fourier number Dt/a^2 with D/a^2 Arrhenius in T. Under vacuum, the driving-force factor $1 - \exp(-\Delta\mu/RT)$ is unity to better than 10^{-9}, so x depends only on the heating schedule. Diffusion and conversion then differ only in the shape of $\alpha(x)$ and in the fitted E_a.

Table 1.4: Kinetic models: reaction model $h(\alpha)$ and conversion as a function of the Arrhenius-weighted time, $\alpha(x)$.

Model	$h(\alpha)$	$\alpha(x)$
R2 contracting cylinder	$2(1-\alpha)^{1/2}$	$1-(1-x)^2$
R3 contracting sphere	$3(1-\alpha)^{2/3}$	$1-(1-x)^3$
F1 first order	$1-\alpha$	$1-e^{-x}$
A2, A3 Avrami ($n = 2, 3$)	$n(1-\alpha)[-\ln(1-\alpha)]^{1-1/n}$	$1-e^{-x^n}$
D3 sphere diffusion	—	$1-\frac{6}{\pi^2}\sum_{m=1}^{\infty}\frac{e^{-m^2\pi^2 x}}{m^2}$

1.4.1 Fits to the vacuum data

Table 1.5 gives the fit of each model to the cumulative KAF ^{3}He release within the fitting window. It also gives the misfit of the resulting out-of-sample prediction of HDM. F1 gives the best fit and the best prediction. R3 and R2 are close behind, so we describe the mechanism as nucleation- or interface-controlled rather than claim first order specifically. The Avrami fits reach the lower bound of E_a and fail on both data sets. Spherical diffusion with $\phi = 0$ fits KAF 4.4 times worse than F1 and cannot produce the $F \approx 0.89$ plateau. To compare diffusion on the same footing, we also fit D3 with ϕ fixed at the F1 value and with ϕ free, over the same 18 KAF steps. The imposed plateau improves the KAF misfit to 0.026 and 0.019, which still ranks below F1, R3 and R2, and the out-of-sample prediction of HDM does not improve (rms 0.056 and 0.052, against 0.018 for F1). These fits also require E_a of 243–289 kJ mol^{-1}. The rate factor for the trap-isolated steps is fixed at $s = 1$ in all three D3 fits, because a diffusive process should not respond to the water pressure at all.

The fitting window excludes the low-temperature steps, where release is dominated by diffusion from goethite, and the plateau, where it is dominated by diffusion from hematite; the rms misfits are computed over the same steps. Repeating the fits with every step included gives the same result: F1 remains the best model ($E_a = 122.7$ kJ mol^{-1}, $\phi = 0.112$, rms 0.0073 for KAF and 0.026 for the HDM prediction), followed by R3 (99.4 kJ mol^{-1}; 0.0097, 0.030) and R2 (87.9 kJ mol^{-1};

0.012, 0.033), with D3 at 177.4 kJ mol^{-1} (0.063, 0.062; 0.022 and 0.042 with ϕ fixed at the F1 value).

Table 1.5: Kinetic and Diffusive Model fits to the vacuum experiments (from Monteiro (2017) and Farley et al. (2024)). All rms values are over the same 18 KAF and 19 HDM steps. D3 is fitted with ϕ fixed at 0, fixed at the F1 value, and free.

Model	E_a	$\ln A$	ϕ	rms KAF	rms HDM
R2	92.8	10.63	0.149	0.0133	0.0291
R3	103.1	12.73	0.136	0.0114	0.0240
F1	120.8	18.15	0.101	0.0087	0.0176
A2	25.5	−4.73	0.000	0.0212	0.1906
A3	20.0[a]	−5.85	0.203	0.0433	0.2342
D3, $\phi = 0$	195.6	32.43	0.000	0.0385	0.0484
D3, $\phi = \phi_{\mathrm{F1}}$	242.8	43.91	0.101	0.0258	0.0557
D3, ϕ free	288.5	55.04	0.184	0.0186	0.0520

E_a in kJ mol^{-1}, A in s^{-1}. [a]At the lower bound of the fit.

1.5 Equilibrium Thermodynamics of Goethite Dehydroxylation

Gibbs energies of the solids. The Gibbs energy of a phase at (T, P) follows from integrating $dG = -S\,dT + V\,dP$ from the reference state $(T_r, P_r) = (298.15\ \mathrm{K}, 1\ \mathrm{bar})$, first in temperature at P_r and then in pressure at T. At P_r, $H(T) = H(T_r) + \int_{T_r}^{T} C_p\,dT$ and $S(T) = S(T_r) + \int_{T_r}^{T} (C_p/T)\,dT$, so $G(T, P_r) = H(T) - TS(T)$. Adding $\int_{P_r}^{P} V\,dP$, and treating the solids as incompressible with no thermal expansion, gives

$$G_i(T,P) = \Delta_f H_i^0 + \int_{T_r}^{T} C_{p,i}\,dT \;-\; T\left[S_i^0 + \int_{T_r}^{T} \frac{C_{p,i}}{T}\,dT\right] \;+\; V_i\,(P - P_r), \tag{1.4}$$

where $H(T_r) = \Delta_f H_i^0$ and $S(T_r) = S_i^0$. This is the "apparent" Gibbs energy of formation (Berman, 1988); the element-entropy terms that distinguish it from the true Gibbs energy of formation cancel in any balanced reaction. Compressibility and thermal expansion change V_i by $\lesssim 1\%$ over the conditions considered and are neglected.

Goethite heat capacity uses the Berman–Brown form (Berman and Brown, 1985) refined by Majzlan et al. (2003), $C_p = k_0 + k_1 T^{-1/2} + k_2 T^{-2} + k_3 T^{-3}$. Hematite uses the same polynomial with the coefficients of Berman (1988), which Majzlan et al. (2003) held fixed in their refinement. Both heat capacities are integrated analytically.

Water. Water is evaluated with the IAPWS-95 equation of state (Wagner and Pruß, 2002), implemented in CoolProp (Bell et al., 2014), which is valid for liquid, vapor and supercritical water to 1 GPa. In the same apparent convention,

$$G_{\mathrm{w}}(T,P) = \Delta_f H_{\mathrm{w}}^0 + [h(T,P) - h(T_r,P_r)] - T\left[S_{\mathrm{w}}^0 + s(T,P) - s(T_r,P_r)\right], \tag{1.5}$$

where h and s are the IAPWS-95 molar enthalpy and entropy and the reference state is liquid water at T_r, P_r. Table 1.6 lists all input data. We use the enthalpies and entropies of Majzlan et al. (2003) refined by mathematical programming against the phase-equilibrium brackets of Voigt and Will (1981).

Table 1.6: Thermodynamic data used for Eq. (**??**).

	Goethite (α-FeOOH)	Hematite (α-Fe_2O_3)
$\Delta_f H^0$ (kJ mol^{-1})	-561.4	-825.8
S^0 (J mol^{-1} K^{-1})	59.5	87.4
V^0 (cm^3 mol^{-1})	20.881	30.274
C_p coefficients	$k_0 = 133.8765$	$k_0 = 146.86$
	$k_1 = -738.5577$	$k_1 = 0$
	$k_2 = -2.177657 \times 10^6$	$k_2 = -5.5768 \times 10^6$
	$k_3 = 2.165124 \times 10^8$	$k_3 = 5.2563 \times 10^8$
C_p(298 K) (J mol^{-1} K^{-1})	74.8	104.0
γ_{hydrated} (J m^{-2})	0.60 ± 0.10	0.75 ± 0.16
$\gamma_{\text{anhydrous}}$ (J m^{-2})	0.91 ± 0.09	1.90 ± 0.30

Enthalpy, entropy, goethite volume and C_p: Majzlan et al. (2003); hematite volume and C_p: Berman (1988); surface energies: Navrotsky et al. (2008). Water: $\Delta_f H^0_{\text{w}} = -285.83$ kJ mol^{-1}, $S^0_{\text{w}} = 69.95$ J mol^{-1} K^{-1} (Cox et al., 1989).

As a check, at 298.15 K and 1 bar with liquid water, Eqs. (**??**)–(1.5) give $\Delta G_r = -0.26$ kJ: hematite plus liquid water is favored by only 0.13 kJ per mole of FeOOH. This is well inside the ± 1.4 kJ mol^{-1} uncertainty that Majzlan et al. (2003) assign to the goethite–hematite free-energy difference; the two assemblages are essentially balanced at ambient conditions. We carry this uncertainty (± 2.8 kJ per reaction as written) as bounding curves on the phase diagram.

Surface energy. A crystal of finite size carries an excess free energy at its surface in addition to the free energy of its bulk (Gibbs, 1878). For a population of particles of fixed size, this excess is γA, where γ is the surface energy (J m^{-2}) and A the total surface area. Both the bulk and surface terms scale with the amount of material at fixed particle size, so the surface contribution can be written per mole, $G = G_{\text{bulk}} + \gamma A_m$, with A_m the surface area per mole. A_m grows as $1/d$, so the term is negligible for coarse crystals and significant for nanoparticles (Navrotsky et al., 2008). For the reaction as written (one mole of hematite from two moles of goethite),

$$\Delta G_{\text{surf}} = \gamma_{\text{hem}} A_{m,\text{hem}} - 2\,\gamma_{\text{goe}} A_{m,\text{goe}}, \tag{1.6}$$

where $A_{m,\text{hem}}$ is per mole of Fe_2O_3 and $A_{m,\text{goe}}$ per mole of FeOOH.

A size must be chosen for each phase. We assume pseudomorphic dehydroxylation, in which hematite inherits the external envelope of the parent goethite, so that both phases share the same external surface area per mole of Fe. This is also the basis of the stability diagrams of Navrotsky et al. (2008). For spherical goethite grains of diameter d this shared area is

$$A_{\text{Fe}} = \frac{6\,V_{\text{goe}}}{d} \quad \text{per mole of Fe.} \tag{1.7}$$

Goethite has one Fe per formula unit, so $A_{m,\text{goe}} = A_{\text{Fe}}$. Hematite has two, so $A_{m,\text{hem}} = 2A_{\text{Fe}}$. Substituting into Eq. (1.6),

$$\Delta G_{\text{surf}} = \gamma_{\text{hem}}\,(2A_{\text{Fe}}) - 2\,\gamma_{\text{goe}}\,A_{\text{Fe}} = 2A_{\text{Fe}}\,(\gamma_{\text{hem}} - \gamma_{\text{goe}})\,. \tag{1.8}$$

The factor of two counts the two moles of Fe that pass through the reaction. It multiplies both terms because both phases carry the same area per Fe. For $d = 10$ nm, $A_{\text{Fe}} = 6 \times 20.881 \times 10^{-6}/10^{-8} = 1.253 \times 10^4$ m^2 mol^{-1}, and with hydrated surface energies $\Delta G_{\text{surf}} = 2 \times 1.253 \times 10^4 \times (0.75 - 0.60) = +3.76$ kJ. The corresponding values are $+0.38$ kJ at 100 nm and $+0.04$ kJ at 1 µm. A positive ΔG_{surf} stabilizes goethite. If the two phases are instead compared at equal particle diameter, the same hydrated surface energies slightly destabilize fine goethite (-1.41 kJ at 10 nm). The choice affects only the stability discussion; the kinetic analysis uses bulk driving forces.

We use hydrated surface energies. They are referenced to liquid water, so they apply wherever the surfaces are in equilibrium with water at or near saturation, which includes all of the hydrothermal experiments. Anhydrous surface energies apply only to surfaces that have lost their chemisorbed water, i.e. under vacuum.

Phase diagram. The phase diagram (Fig. **??**) is computed by solving $\Delta G_r(T, P) = 0$ (Eqs. (**??**)–(1.8)) numerically for $P = P_{\mathrm{H_2O}}$ at each temperature, using Brent's method on $\log_{10} P$ between 10^{-12} and 10^4 bar, with the solids at the fluid pressure. Where no root exists below the melting curve of ice, the curve is truncated.

Driving force and the role of water. The quantity that matters for kinetics is the driving force for hematite formation, $\Delta\mu \equiv -\Delta G_r/N_A$ (per mole of Fe_2O_3). Water controls $\Delta\mu$ through its own chemical potential, because it is a product of reaction (**??**). G_{w} in Eq. (**??**) is the chemical potential of H_2O at the prevailing water fugacity. It depends on the partial pressure of water, not on the total pressure. An inert gas, or any other mechanical load, acts only through the solid-volume term $\Delta V_s(P - P_r)$, which contributes $-1.15\ \mathrm{J\,bar^{-1}}$ and slightly favors the denser hematite. At 40 bar that is 46 J, three orders of magnitude smaller than the effect of 40 bar of water. The water-buffered step-heating technique therefore stabilizes goethite primarily by reducing the driving force of the reaction. Surface hydration is a second-order correction that matters only for nanometer grains (Eq. (1.8)).

1.6 Multiple Diffusion Domain and Thermal History Modeling

1.6.1 Forward modeling radiogenic ^{4}He in nature and in the laboratory

Each diffusion domain produces ^{4}He uniformly at rate P and loses it by diffusion through a zero-concentration boundary from the time of crystallization, t_{form}. The concentration is expanded in the eigenfunctions of the sphere, $C = \sum_k A_k(t)\sin(k\pi x)/x$, where x is radial position normalized to the domain radius and A_k is the amplitude of the k-th mode (Carslaw and Jaeger, 1959). Each mode amplitude A_k evolves as

$$\frac{\mathrm{d}A_k}{\mathrm{d}t} = -k^2\pi^2\frac{D}{a^2}A_k + c_k P, \qquad c_k = \frac{2(-1)^{k+1}}{k\pi}, \tag{1.9}$$

where the amplitude A_k of each mode k decays by diffusion and grows by production P. For the initial condition, at $t = 0$, all amplitudes are zero. Then, for a discrete timestep Δt at constant temperature, each mode is updated as

$$A_k \leftarrow A_k e^{-\lambda_k} + c_k P\,\Delta t\,\frac{1 - e^{-\lambda_k}}{\lambda_k}, \qquad \lambda_k = k^2\pi^2\frac{D}{a^2}\Delta t. \tag{1.10}$$

Because the eigenfunctions do not depend on D, the present-day amplitudes $A_{k,j}$ carry each domain's non-uniform natural ^{4}He profile into the laboratory, where each mode decays as $\exp(-k^2\pi^2\tau)$. The ^{4}He remaining in domain j when the reference domain has reached laboratory time τ is therefore

$$N_j(\tau) = 3\sum_k A_{k,j}\,\frac{(-1)^{k+1}}{k\pi}\exp\left[-k^2\pi^2\tau\left(\frac{r_j}{r_0}\right)^{-2}\right], \tag{1.11}$$

and the predicted step ratio is

$$\frac{R_{\mathrm{step},i}}{R_{\mathrm{bulk}}} = \frac{\Delta N_{4,i}/N_{4,\mathrm{tot}}}{\Delta F_i^{\mathrm{obs}}}, \qquad \Delta N_{4,i} = \sum_j f_j\left[N_j(\tau_{0,i-1}) - N_j(\tau_{0,i})\right], \tag{1.12}$$

where $N_{4,\mathrm{tot}} = \sum_j f_j N_j(0)$ includes the gas released at fusion. The reference times $\tau_{0,i}$ are solved from the *measured* cumulative ^{3}He by inverting Eq. (**??**), so the prediction does not depend on the heating schedule or on step-to-step scatter in the kinetics, and no uniform ^{4}He distribution is assumed at the start of the experiment. Taking U and Th to be distributed like the ^{3}He, the bulk He age is $t_{\mathrm{He}} = \sum_j f_j N_j(0)/P$.

1.6.2 Multiple diffusion domain fits for $N = 1$–**5**

Figure 1.7 shows the fitted fractional and cumulative ^{3}He yields, and the Arrhenius and delta plots they imply, for one to five domains. The fits for all N are summarized in Table **??** of the main text.

1.6.3 Two-domain alternative

The two-domain structure (2.2% of the ^{3}He at $r/r_0 = 5.4 \times 10^{-3}$, 97.8% at 1.05) fits the ^{4}He/^{3}He spectrum better than the adopted four-domain structure: a burial path with crystallization at 195 Ma and a peak of ∼66 °C at ∼100 Ma gives $\chi^2_\nu \approx 1.3$. However, this structure reproduces the ^{3}He Arrhenius array poorly ($\chi^2_\nu = 3.3$; Fig. 1.7), so we do not adopt it.

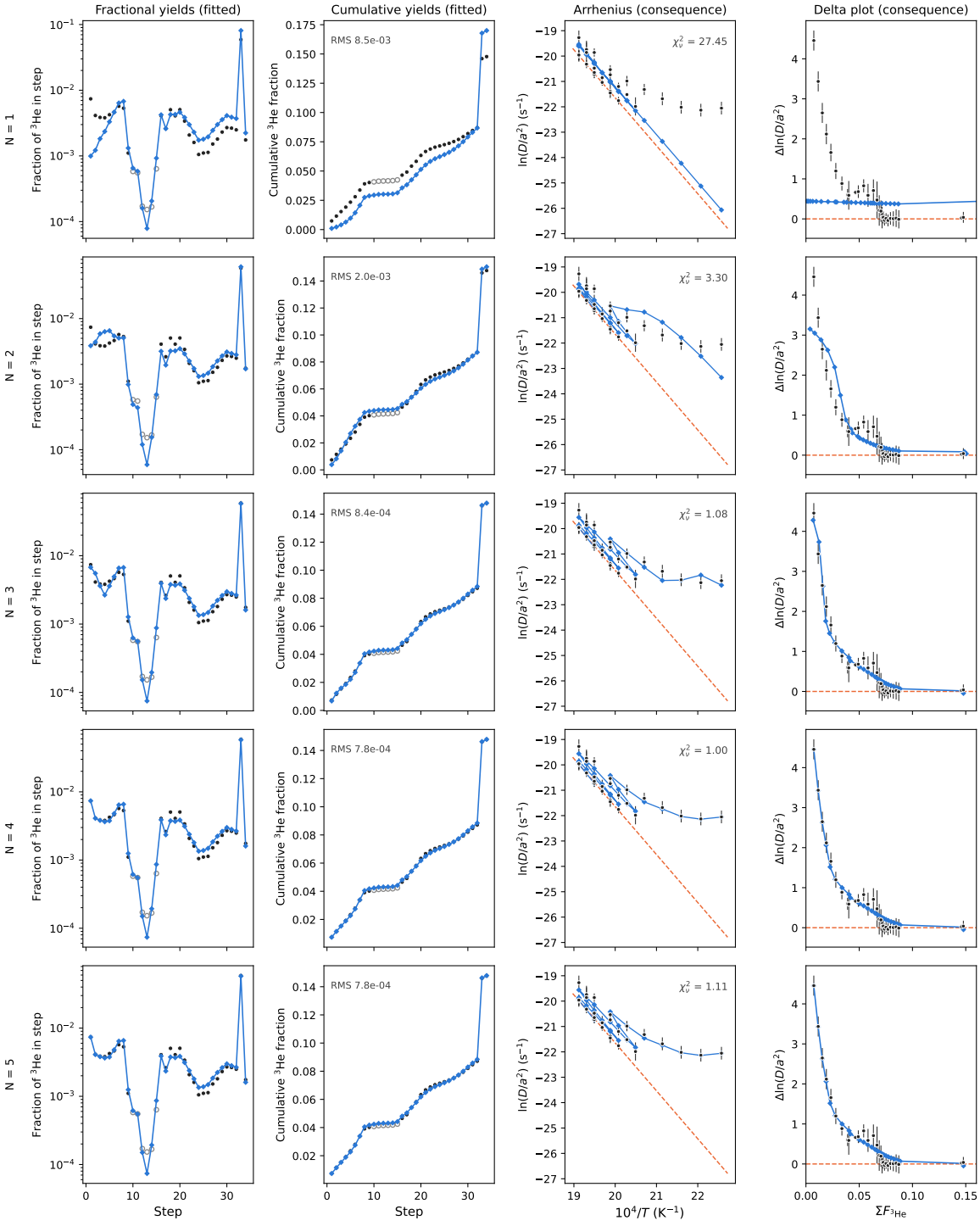


Figure 1.7: Multiple diffusion domain fits to the ^{3}He release of HT-Win-06-01b for $N = 1$–5 domains (rows). Columns: fitted fractional and cumulative yields, and the Arrhenius and delta plots implied by each fit. Hollow grey symbols are the six low-signal steps, which are not fitted and are omitted from the Arrhenius and delta plots.

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